

Supplementary Methods

Hikaru Wakaura

QIRI (Quantum Integrated Research Institute Inc.), Tokyo 107-0061, Japan

E-mail: h.wakaura@qiri.co.jp

Abstract. Model definitions, additional numerical results, probability-bound derivations, evidence scope and reproduction instructions for “Conditions for transmitting cryptochrome spin-state history to neural responses”.

S1. Molecular maps and provenance

The retained nuclei are flavin N5 and N10, both spin-1 ^{14}N . The closed-shell nuclear density matrix has dimension nine. The escape-state electron–nuclear density matrix has dimension 18; it is not assumed to close on nine nuclear populations. Vectorisation is column-major. The supplied archive contains the state Hamiltonians, dissipators and reaction maps in `latest_case.npz`, and the evaluated reference cycle and prepared states in `modes_and_states.npz`. These frozen operators are the numerical starting point for the supplied independent reproduction. It does not recompute their upstream electronic-structure or molecular-dynamics inputs.

The waiting-state generators have the Lindblad form [23]

$$L\rho = -i[H, \rho] + \sum_{\alpha} \kappa_{\alpha} \left(A_{\alpha} \rho A_{\alpha}^{\dagger} - \frac{1}{2} \{ A_{\alpha}^{\dagger} A_{\alpha}, \rho \} \right). \quad (\text{S1})$$

The implementation constructs non-negative κ_{α} and Hermitian noise operators by diagonalising a finite-window covariance estimate. This yields a unital, trace-preserving waiting model. It is not a complete Bloch–Redfield treatment of measured fluctuations. The MD-based electron-relaxation calculations in [24] are relevant methodological precedent, not a source of our nuclear relaxation constants.

For the radical-pair block, the loss added to this generator is

$$L_{\text{loss}}\rho = -\frac{1}{2}\{k_S P_S + k_E I, \rho\}. \quad (\text{S2})$$

The product and escape maps integrate the corresponding k_S singlet projection and k_E escape flux. This is a Haberkorn-type reaction construction [21, 22]. The derivation in [22] also identifies reactive exchange and higher-order singlet–triplet dephasing corrections; they are not included in the frozen reference operators. Thus naming the standard reaction formalism does not establish that all reaction-induced spin effects have been retained.

For a generator L and exponentially distributed waiting time of rate r , the waiting map is

$$W(L, r) = \int_0^\infty r \exp(-rt) \exp(Lt) dt = r(rI - L)^{-1}. \quad (\text{S3})$$

The cycle in the reference implementation is

$$M = W_H(b)W_R(a) [W_R(c_P)P + T_E W_E(c_E)E]. \quad (\text{S4})$$

Here P maps the input nuclear state to the product-branch nuclear output, E maps it to the electron-containing escape output, and T_E traces out the escape electron. The product branch and subsequent reduced closed-shell state use the adopted R generator; the HQ wait uses the H generator. The reference rates are $a = b = c_P = 10 \text{ s}^{-1}$ and $c_E = 4 \text{ s}^{-1}$. Reaction maps integrate spin-selective reaction and escape with rates 10^7 and $4 \times 10^6 \text{ s}^{-1}$. All tensor and noise coefficients used by this frozen model are recorded in the operators, rather than reconstructed from a literature lifetime.

Writing $x_0 = \text{vec}(I_9/9)$ and r_P for the product-yield row, the three reported yields are obtained from $r_P M^{20} x_0$, $r_P (D_P M D_P)^{20} x_0$ and $r_P (R_0 M R_0)^{20} x_0$, where $D_P = C C^\dagger$ projects onto HQ-basis populations and $R_0 = x_0 \text{vec}(I_9)^\dagger$ resets the nuclei. The supplied reproduction evaluates these expressions independently from the stored matrices. The reference magnetic field is $50 \mu\text{T}$ and additional oxygen-spin relaxation is zero. The archive also contains some non-reference interventions; their presence does not imply that they are physiological estimates.

The adopted 6 ps oxidised-state trajectory and 1 ps noise window do not measure long-time fluctuations in all chemical states. Fixed molecular orientation, chemical-state transfer of noise information, a reduced noise approximation and long-time Markovian propagation remain assumptions. This limits both the numerical relaxation times and the physical range over which the population reduction can be interpreted.

S2. Population modes, renewal and counterexamples

In the adopted HQ basis, $G = C^\dagger D C$ is a symmetric, uniform-stationary population generator within numerical tolerance. Its eight non-normalisation modes have decay rates λ_k . The sign convention for eigenvectors changes the write amplitude and read sensitivity together, leaving their product invariant. Table S1 records all eight modes. Modes 1 and 4 are N10-local, 2 and 5 are N5-local, and 3, 6, 7 and 8 are diagonal correlation modes. The isolated half-decay time is the root of $\sum_k a_k b_k \exp(-\lambda_k t) = \sum_k a_k b_k / 2$. It is not the longest mode time constant. The 43.41% correlation share is a linear-mode readout decomposition; replacing a correlated distribution by the product of its marginals defines a different finite-change diagnostic. The diagonal correlations do not establish entanglement.

The complete sensitivity table has 216 rows: full, population and reset at 72 settings. Reference one-factor counterexamples include $c_P = 0.1 \text{ s}^{-1}$, $c_E = 10^4 \text{ s}^{-1}$ and a 100-fold closed-shell noise multiplier. The range is an assumed numerical

Table S1. All eight population modes in the reference preparation. Rates are in s^{-1} , time constants in seconds and shares in percent.

Mode	λ_k	$1/\lambda_k$	a_k	b_k	Share
1	0.067455	14.8248	0.00613574	0.0540324	5.8143
2	0.135948	7.3557	0.000612594	0.00700285	0.0752
3	0.203403	4.9163	-0.000419867	-0.00526286	0.0388
4	0.295818	3.3805	-0.0159478	-0.145395	40.6659
5	0.409168	2.4440	-0.00724261	-0.0789909	10.0335
6	0.431766	2.3161	0.00111469	0.0143781	0.2811
7	0.476623	2.0981	0.0050696	0.0583887	5.1914
8	0.704986	1.4185	-0.0134699	-0.160434	37.9000

exploration. At fixed short phase-preserving waits, the saved coherence contribution reaches approximately 4.85×10^{-6} . Removing the HQ energy gap in a mathematical control produces a full-Pauli difference near 2.09×10^{-4} after 10 s. Setting all chemical wait rates to 10^8 s^{-1} produces a full-population difference near 2.89×10^{-4} , but compromises the assumed separation of reaction and wait times.

The saved HQ population-coherence coupling Frobenius norm is about 0.621 s^{-1} and the minimum energy gap about $4.80 \times 10^5 \text{ rad s}^{-1}$. Fast phase rotation and wait averaging can support a small correction. Their ratio is not, by itself, a proved error bound on the entire reactive readout. An error criterion that predicts the observed failures in withheld physical conditions has not been established.

S3. Chemical yields and redox-site accounting

In the equal-endpoint example, P immediately supplies one peroxide equivalent; E supplies one superoxide at escape and another on complete reoxidation. Complete dismutation gives the same final mean peroxide yield in both branches. If removal has rate κ and reoxidation rate k_E , the difference in retained peroxide obeys

$$\dot{h} = -\kappa h - \frac{\Delta q k_E}{2} \exp(-k_E t), \quad h(0) = \frac{\Delta q}{2}. \quad (\text{S5})$$

For $\kappa \neq k_E$,

$$h(t) = \frac{\Delta q}{2} \left[\exp(-\kappa t) - \frac{k_E}{\kappa - k_E} \{ \exp(-k_E t) - \exp(-\kappa t) \} \right]. \quad (\text{S6})$$

This expression reproduces the positive 200 ms and negative 2 s residuals at $k_E = 1$, $\kappa = 5 \text{ s}^{-1}$. If delivery fractions are $\eta_P, \eta_{E,e}, \eta_{E,l}$ for P, early E and late E, the final delivered difference becomes

$$\Delta q \left[\eta_P - \frac{\eta_{E,e} + \eta_{E,l}}{2} \right]. \quad (\text{S7})$$

Thus equal endpoints are a counterexample to a universal yield-to-total-product map, not proof that every local response vanishes. Light-driven peroxide and cellular indicator

observations [10] do not supply the dark-branch delivery fractions. Nor is miniSOG singlet-oxygen input interchangeable with a dark-CRY oxygen-species input.

For the three Hk states in the main text, the mean first-passage time from O to R is $(h + o)/(hb) + 1/h$, when $b, h > 0$. This is not generally the single-exponential lifetime of total oxidised occupancy. A rate fitted during sustained substrate exposure cannot be substituted for the cofactor-off rate without specifying the chemical model.

The exponential writing-pulse calculations assume continuous supply implicit in $a(t)$. Their total oxidation peaks should not be divided by the nuclear yield difference to claim a biological amplification factor. The $1 \mu\text{M}$, $0.01 \mu\text{m}^3$ substrate example gives a mean count, not exactly six molecules in every realisation. With initially reduced sites and one-use substrate, mean new oxidation is bounded by the mean initial count. The 200-site example therefore tests material sufficiency independently of reaction speed. Membrane site number is also distinct from a freely dissolved concentration.

S4. Circuit dynamics and stored controls

The source consists of N two-state slots. At the cue, the two counts are coupled binomial samples at q_0 and q_1 . Between packets separated by $\Delta t = 10$ ms, each arm recovers towards q_0 with

$$P(1 | 0) = q_0(1 - d), \quad P(1 | 1) = d + q_0(1 - d). \quad (\text{S8})$$

Here $d = \exp(-\Delta t/\tau)$. The source trajectories are identical before the cue. Until chemical arrival the circuit reads the reference trajectory in both arms, so pre-arrival background history is correlated with the reference prepared count. This point matters for the bounds in section S6.

Heavy-tailed strengths have been measured among excitatory connections in rat visual cortex [39]. This motivates testing heterogeneous weights, but does not determine the signed synthetic network here or a CRY-sensitive circuit. Network weights start from lognormal draws with log standard deviation 1.2, are rescaled to total unsigned weight 0.45 before independent sign assignment, and are made negative with probability 0.25. The main architecture seed is 26091681. Delays are discrete draws from 0, 10, 20 and 30 ms with architecture seed plus 2. Random allocation uses architecture seed plus 1; top allocation sorts $w_j \exp[-d_j/(40 \text{ ms})]$ before trial outcomes. Shared chemical noise already correlates releases even when the additional shared-uniform release parameter is zero, as in the main case.

At each packet, independent route-specific release uniforms generate centred releases $r_j - p_0$. Equal voting uses coefficients $1/[8\sqrt{p_0(1-p_0)}]$; weighted input uses $w_j/[\|w\|_2\sqrt{p_0(1-p_0)}]$. Delayed inputs are queued. Temporal integration uses $x_{t+1} = d_I x_t + (1 - d_I)e_t$, $d_I = \exp[-\Delta t/(40 \text{ ms})]$, and divides the readout by $\sqrt{(1 - d_I)/(1 + d_I)}$. The reference threshold is $(32 - 64p_0)/[8\sqrt{p_0(1-p_0)}]$. Fixed per-trial, route-specific normal offsets have standard deviation 0.15. The competition variant adds simultaneous inhibition with coefficient 0.3 between go and stop after triggering.

Independent-race and interactive-race accounts are established precedents [18, 19]; the present discretisation, threshold and inhibition coefficient are independently assumed. Earliest eligible go time is 350 ms; trigger and stop timing are as in the main text. The full event rules are in the archived implementation.

The main paired confirmation uses trial seeds 26091711, 26091712 and 26091713 with 8192 pairs each. All 1932 summary rows, the 432-row grid and 540 additional-control rows are supplied. Rows are model outputs, not biological replicates. Zero gain, reset, blockade, delayed arrival, opposite sign, background changes, depletion and replay controls are included. Primary confirmation event times are archived for independent reclassification. Stored grid intervals are not substituted for the higher-trial confirmation intervals.

Matching the total probability modulation does not match weighted mean or covariance. Top-eight performance therefore cannot isolate sparsity from readout alignment. The non-maintained-background control also moves the operating point: selected connections approach release probability 0.0225 while non-selected connections stay at 0.45. It does not change only the lifetime. Replaying an identical source into identical downstream rules tests reproducibility of the downstream map, not a separately derived classical upstream mechanism.

At preparation, the nonlinear transformation places approximately 24.54% of reset and 57.39% of full count draws beyond 90% of the saturation width at $g = 1024$. These are initial-distribution properties, not all-time circuit statistics. The mean initial per-selected-connection contrast is about 0.31093, which differs from evaluating the nonlinear function at the mean count. The finite-gain plateau reported in an external model review lacks an independently verified execution record and is not included as a result.

S5. Effective fit, prediction and identifiability

The source dataset comprises 11 cells from figure 4(f) and 22 from figure 6(d) of [9]. Repeated presentation of the same measurement is not counted as a new cell. Odd within-figure cell numbers define fitting cells, even numbers define held-out cells. All 40 min measurements are excluded from fitting; seven held-out cells contribute to the 40 min validation. Missing observations are retained in the ledger without imputation. Animal/batch identifiers are unavailable.

For component $m \in \{\text{fast}, \text{slow}\}$, define $y_m(t) = \log[\tau_m(t)/\tau_m(0)]$. The effective coordinate x follows

$$\begin{aligned} \dot{x} &= k(1 - x), & x(0) = 0 & \text{ outside the train,} \\ \dot{x} &= k(1 - x) - v(x + B) & \text{ during the 10–30 min train,} \\ \hat{y}_m(t) &= A_m x(t). \end{aligned} \tag{S9}$$

Rates are in min^{-1} . The voltage-train rate is an effective average over the applied protocol, not the microscopic exchange rate at one voltage. The five dynamic parameters

are $k, v, B, A_{\text{fast}}, A_{\text{slow}}$. The static model predicts a fitted constant for each component at all positive observation times and zero at baseline. The no-voltage model sets $v = B = 0$ and fits k and the two amplitudes.

Least squares uses residual weights $1/[s_m\sqrt{n_{im}}]$, where n_{im} is the number of available observations for cell i , component m , and s_m is its component’s training-data standard deviation. The stored scales are 0.351667 and 0.727282. The dynamic search uses $10^{-4} \leq k \leq 10$, $10^{-4} \leq v \leq 100$, $0 \leq B \leq 10$ and $0 \leq A_m \leq 4$. Five starts are used; rate coordinates are logarithmic. The no-voltage model uses three starts. The optimiser is bounded least squares with the saved implementation’s tolerances of 10^{-10} and maximum 1500 function evaluations.

The principal dynamic fit has $k = 0.0514559$, $v = 0.0978634 \text{ min}^{-1}$, $B = 0.838721$, $A_{\text{fast}} = 0.601116$ and $A_{\text{slow}} = 1.312229$. These are effective observation-model estimates. The resampling follows the bootstrap framework [40]. To preserve the observed within-cell structure, the 200 training-cell bootstrap fits resample whole cells, keeping their components and times together; the upper-amplitude-bound sensitivity check refits the same resamples with the bound increased to 8. Stored held-out cell-bootstrap percentile intervals use 4000 resamples (seed 2026091742) and concern prediction-error differences and are distinct from the parameter bootstrap. None models unknown animal/batch dependence.

Structural non-identifiability and finite-data uncertainty must be distinguished [41]. We demonstrate the scale symmetry directly; we do not perform the profile-likelihood analysis proposed in that reference. An explicit family illustrates the scale ambiguity. For an admissible excursion $D > 0$, let $z_0 = BD$, $z_{\text{rest}} = (B + 1)D$ and the observation gain be A_m/D . Then $\hat{y}_m = (A_m/D)(z - z_0)$ is unchanged when D changes with corresponding rates in the admissible family. Absolute occupancy is not identified by the observed product. The supplied table records several such realisations. At the no-voltage optimum $k = 3.175914 \text{ min}^{-1}$, $\exp(-10k) \simeq 1.61 \times 10^{-14}$, explaining the near-static prediction at the first nonzero observation time without implying a duplicated-code error.

The original finite-miniSOG experiment (figure 4(c,d) and “A redox memory of lipid peroxidation” in [9]) provides evidence for persistence after finite input. It is a different evidential claim from the predictive comparison of the sustained-input subset here. Lack of dynamic predictive superiority neither falsifies chemical storage nor establishes equivalence of mechanisms.

S6. Preparation and readout bounds

S6.1. Equality-preserving paired preparation

The coupling inequality and dual definition of total variation are standard [28], section 2, “Total variation distance”. The following proof states their application to the actual intervention. For $q_1 \geq q_0$, define $A = \sum_{i=1}^N 1(U_i < q_0)$ and $B = \sum_{i=1}^N 1(U_i < q_1)$ with

independent uniform U_i . Then

$$P(A \neq B) = 1 - (1 - q_1 + q_0)^N. \quad (\text{S10})$$

Equivalently, conditional on A , the excess is binomial with $N - A$ trials and probability $(q_1 - q_0)/(1 - q_0)$. The implementation uses this equivalent count coupling. If equal counts at the cue yield equal future counts, releases and outcomes under common randomness, then $|E[S_1 - S_0]| \leq E[|S_1 - S_0|] \leq P(A \neq B)$. The same expression holds after exchanging labels when $q_0 > q_1$.

This bound uses the actual intervention definition and coupling. It excludes repeated intervention-dependent source resets, arm-specific downstream laws and deterministic-mean source replacement. A bound on the true probability contrast should not be applied as an exact bound on a finite-sample estimate. The $N \geq 27$ example is necessary only for the stated target probability difference and source contrast, and is not a biological molecule-count estimate.

S6.2. A stronger bound requires a common single-count readout

If all intervention dependence enters through a single count n and both arms share $f(n) = P(S = 1 | n)$ with $0 \leq f \leq 1$, then

$$\begin{aligned} |P_1(S = 1) - P_0(S = 1)| &= \left| \sum_n f(n)[p_1(n) - p_0(n)] \right| \\ &\leq \text{TV}(p_1, p_0) = \frac{1}{2} \sum_n |p_1(n) - p_0(n)|. \end{aligned} \quad (\text{S11})$$

This is the usual contraction under a common stochastic readout. Correlated side information requires the joint input distribution instead of its count marginal. For example, let A, B have the monotone preparation coupling. The two observed inputs (A, A) and (A, B) can differ on the event that the coordinates are unequal even when the marginal count distributions are close. At $N = 4$ and the present probabilities, that event difference is 0.0226134, exceeding the marginal total variation 0.00996336. The pre-arrival reference history in M4 prevents an automatic identification of its full readout with the simpler kernel. Both expressions are standard probability bounds specialised to the model. They do not calibrate downstream gain or prove that their upper limits are achievable by the circuit. The supplied script checks the binomial probabilities with a direct probability-mass evaluation and an independent recurrence.

S7. Prior work and evidence scope

The most direct population-inheritance precedent is Wong *et al.* [16]. Their equation (8), section II B, printed page 035102-3 removes all coherences in the Zeeman basis between excitations. Section III D and figure 4, printed pages 035102-7 and 035102-8, compare retained nuclei, radical configurations and branch-specific retention. Section IV, page 035102-9, concludes against substantial compass-performance improvement in

Table S2. Conditional upper bounds in percentage points at $\delta = 0.005701932453$. The single-count column requires an additional common-readout assumption and is not a bound on the full history circuit.

N	Single-count TV	Paired-preparation bound
1	0.570193	0.570193
6	1.367723	3.372761
10	1.729531	5.557831
26	2.727810	13.815041
27	2.794178	14.306462
100	5.308899	43.550573
1,000	16.692003	99.671455
10,000	49.484371	100.000000

the realised model. The four-nucleus S-retention cases nevertheless show a reported 50–100% relative anisotropy enhancement. A relative increase in a small anisotropy is not an equal increase in total chemical output or neural activity.

The preparation of nuclear populations has experimental precedents in a flavin–tryptophan dyad [13], figure 4, and engineered LOV proteins [14], Results and Discussion. Neither measures the N5/N10 storage time assumed here. The structural response in [31], figure 4, and its magnetic-field null in supplementary figure S4 concern a different stored variable. Likewise, [32], figure 2(i) and accompanying Results, attributes slow optical contrast to chemical kinetics, not seconds of spin coherence. These distinctions prevent the reuse of one kind of memory lifetime as another.

The present model differs in its dark cycle, HQ projection basis, selected nuclei, interaction assumptions and observable. Those differences identify a comparison to perform; they are not proof of novelty. A matched-limit reproduction of Wong’s model has not been carried out. The article’s supplementary calculations have not been exhaustively reviewed. Eisenreich *et al.* [12] is cited only for the experiment and interpretation reported in its original abstract; Matysik *et al.* [15], section 2.2, supplies review context rather than a substitute for unchecked original studies.

Table S3 places the proposed Hk route beside other CRY-related transduction observations. These pathways provide comparisons and potential confounds, not interchangeable gains. In [42], section 2.9 specifies fixation before FRET imaging. Protein association is not evidence of stimulus-dependent signal transmission. Similarly, the cofactor-oxidation experiments in [6], Results, motivate channel modulation but do not determine the three-state model’s rates. The transport/removal discussion in [11] motivates compartment-specific measurements rather than a universal peroxide gain.

The QRC comparison uses the driven-dynamics and trained-readout framework of [29], section II. Wakaura and Tanimae [30] is a Research Square preprint, version 1, cited here in preprint form and not used as experimental validation of a biological mechanism. Its sections III–V concern capacity, readout restrictions and a reservoir

Table S3. Published routes constrain different steps and do not establish one calibrated dark-spin-to-synapse chain.

Route	Observed or modelled support	Unresolved for the present hypothesis
CRY-Hk	Fly neuronal light response and Hk rescue [4, 7, 8]; cofactor-current coupling in mammalian Kv β 1 [6]	Dark nuclear-history input, local chemical delivery and presynaptic release gain
CRY-ROS-TRPA1	Antennal support-cell/olfactory pathway under short-wavelength light [35]	Support-cell to neuron mechanism cannot be identified with central Hk or dark-brain stopping
CRY-D1/Gs	Mouse NAc perturbations and heterologous signalling assays [33, 34]	Spin-state dependence is unmeasured; clock and receptor mechanisms provide alternatives
CRY4-G protein	Purified-protein binding and heterologous-cell FRET [42]	No CRY-driven GTP-exchange or neuronal-output measurement in that study

memory horizon; the present cycle-boundary storage observable is different. We do not transfer its capacity values or simulation outputs into this article.

The following evidence labels are restricted to each stated proposition. “Strongly Supported” does not assert a complete biological pathway.

Strongly Supported: CRY-dependent neuronal blue-light responses and redox-sensitive Hk rescue are reported in [4], figure 2(D,E), and [7], figure 3. Light-driven CRY peroxide/indicator responses are reported in [10], figures 3 and 4. Finite-input redox memory is supported by the stated experiments in [9], figure 4(c,d). Population inheritance is explicit in [16], equation (8). These are distinct experimental systems or theoretical claims.

Partially Supported: A CRY-oxygen species-lipid substrate-Hk route combines evidence from [7, 9, 10], but their intermediates, localisation and rates are not jointly measured. The assumed three-state reduction and synaptic gain are not fitted by these references.

Contradicted: Describing the reported seconds-scale nuclear relaxation as an in-vivo measurement contradicts its computational provenance. Describing total Hk oxidation as the full-reset difference contradicts the input definitions. The strong interpretation that Wong’s realistic configurations eliminate every population-history effect conflicts with section III D and figure 4.

Not Found in Uploaded Sources: A calibrated dark-CRY-to-behavioural-stopping effect, a CRY origin for human readiness potentials, or a demonstrated quantum computational advantage. The present source corpus and calculations do not establish these propositions.

For readiness-potential interpretation, [37], abstract and figure 1, supports the existence and evaluation of an accumulation-based account. The sections on direct evidence and the necessity of random fluctuations in [38] challenge a unique microscopic interpretation. For stopping, [20], boxes 1 and 2 and recommendations 2 and 7, motivates attention to triggering and detection; it does not determine the assumed 40/70 ms circuit timings. Combining supportive and challenging citations here guards against inferring a specific molecular cause from an averaged neural or behavioural output.

S8. Supplementary Data and reproduction

A separate source-code and compiled-PDF repository is designated for release at <https://github.com/deeptell-inc/cryptochrome-neural-history>; public access is pending. It excludes simulation outputs, frozen operators and large datasets. The Supplementary Data archive described below is a separate submission artifact and supplies the frozen inputs for the bounded reproduction.

The Supplementary Data archive includes a README, file hashes, figure source tables, frozen operators, primary confirmation event times, fit outputs and a bounded reproduction script. Its numerical paths preserve the module names used during analysis. The original implementation files are included for inspection; the standalone reproduction is the supported entry point for regenerating the article figures and checking the reported summaries from frozen inputs. This is not a complete recreation of all upstream simulations or electronic-structure calculations.

Run `python reproduce.py` from the extracted archive with NumPy, SciPy, pandas and Matplotlib installed. Matrix functions, ODE integration, least-squares optimisation and scalar root finding use SciPy [43]; this citation credits software, not biological validity. It checks the three cycle-boundary yields, the mode sum and lifetime, chemical accounting, selected stopping classifications, Hk error summaries and conditional bounds, and produces the five figures and a JSON check report. The figure data represent existing simulation series; any arithmetic or replotting performed during manuscript preparation does not add an independent experimental replicate.

Stored diagnostics include molecular solver disagreement of approximately 2.7×10^{-10} , Hk finite-input DOP853–Radau difference of 1.97×10^{-11} , tangent-versus-finite-difference discrepancy of 7.14×10^{-10} and chemical ODE agreement near 8.98×10^{-13} . These numerical checks do not validate the assumed biological mechanisms. A previous small-circuit audit recorded runtime warnings in two matrix products; an algebraically equivalent in-memory `einsum` implementation reproduced event times and finite diagnostics. The environment-level cause was not established. The present manuscript preparation does not rerun the full circuit simulation through that warning-producing path.

The essential numerical summaries and mechanistic limits are in the main article; supplementary records document derivations and reproducibility rather than supply an

otherwise absent central result.

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